

Fuel Consumption Prediction (City & Highway)

A Complete End-to-End Data Science Project

Date: 2025-12-27

Deployment: <https://fuel-consumption-calc.streamlit.app/>

Dataset: C:\Users\user\Downloads\Models\Models\vehicles.csv

Notebook reference: C:\Users\user\Downloads\Models\Models\Fuel_Consumption_Models_and_Merging (2).ipynb

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1. Executive Summary

This report presents a complete end-to-end workflow for predicting vehicle fuel consumption in two driving contexts—City and Highway—measured in L/100 km. The pipeline covers data preparation, exploratory analysis, model development, evaluation, and deployment via a Streamlit application.

Project Phases

- Data collection: load a FuelEconomy-style vehicles dataset (vehicles.csv).
- Preprocessing: filter years, normalize fuel types, convert MPG to L/100 km.
- Modeling: evaluate multiple regressors to predict City and Highway jointly.
- Deployment: expose the best-performing models via Streamlit.

2. Project Overview and Objectives

2.1 Problem Statement

Fuel consumption affects total cost of ownership, emissions, and vehicle selection decisions. Given a compact set of vehicle specifications, the objective is to estimate expected fuel consumption for City and Highway driving conditions.

2.2 Project Objectives

- Build a consistent feature pipeline aligned with the deployed app.
- Perform EDA to understand distributions and relationships.
- Train and compare multiple ML models.
- Report clear metrics (MAE, RMSE, R^2) for both targets.
- Provide a deployment-ready inference interface in Streamlit.

3. Data Collection

3.1 Dataset Source

The project uses a FuelEconomy-style vehicles dataset (vehicles.csv) containing vehicle specifications and fuel economy measurements. The generator supports auto-detecting vehicles.csv in the project folder.

3.2 Dataset Overview

Rows (raw)	49529
Rows (filtered)	30215
Columns (raw)	84
Target columns	City (L/100 km), Highway (L/100 km)

4. Data Preprocessing and Cleaning

4.1 Key Transformations

- Filter to modern vehicles (year ≥ 2000).
- Normalize fuel type values into consistent categories.
- Exclude Hydrogen entries to align with deployed feature set.
- Convert MPG to L/100 km using $235.214583 / \text{MPG}$.
- Drop rows with missing values in features/targets.

5. Exploratory Data Analysis (EDA)

This section summarizes key data characteristics relevant to model performance. Figures are generated automatically from the provided dataset.

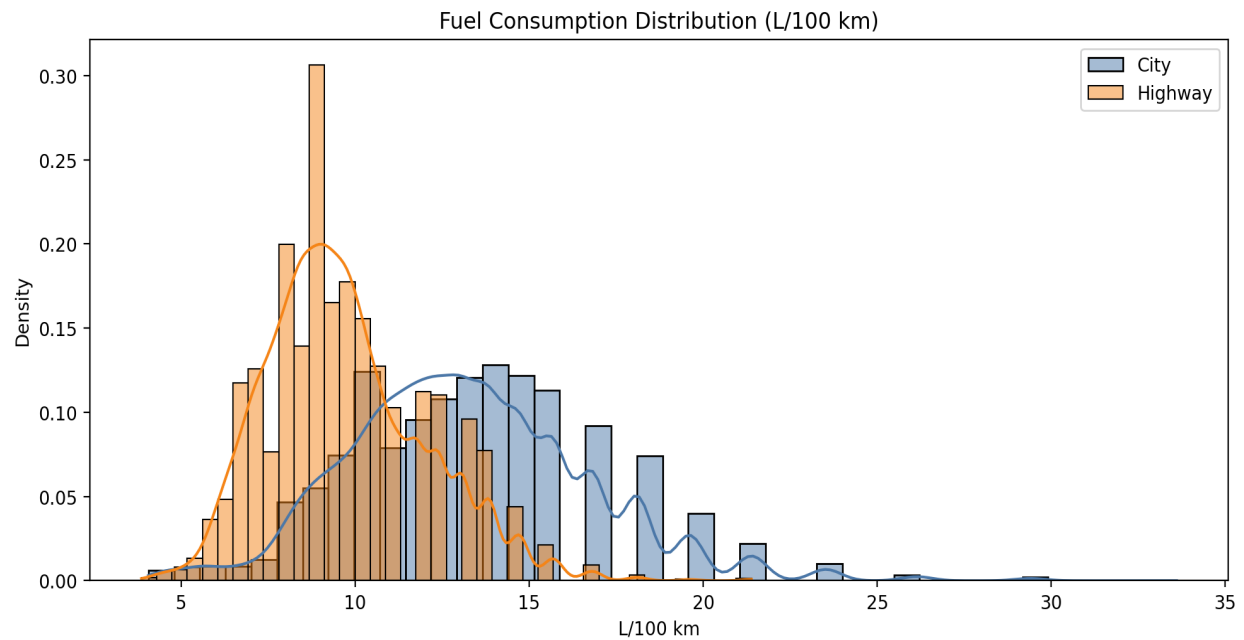


Figure: Histogram + KDE of target distributions after filtering (year ≥ 2000 ; Hydrogen removed).

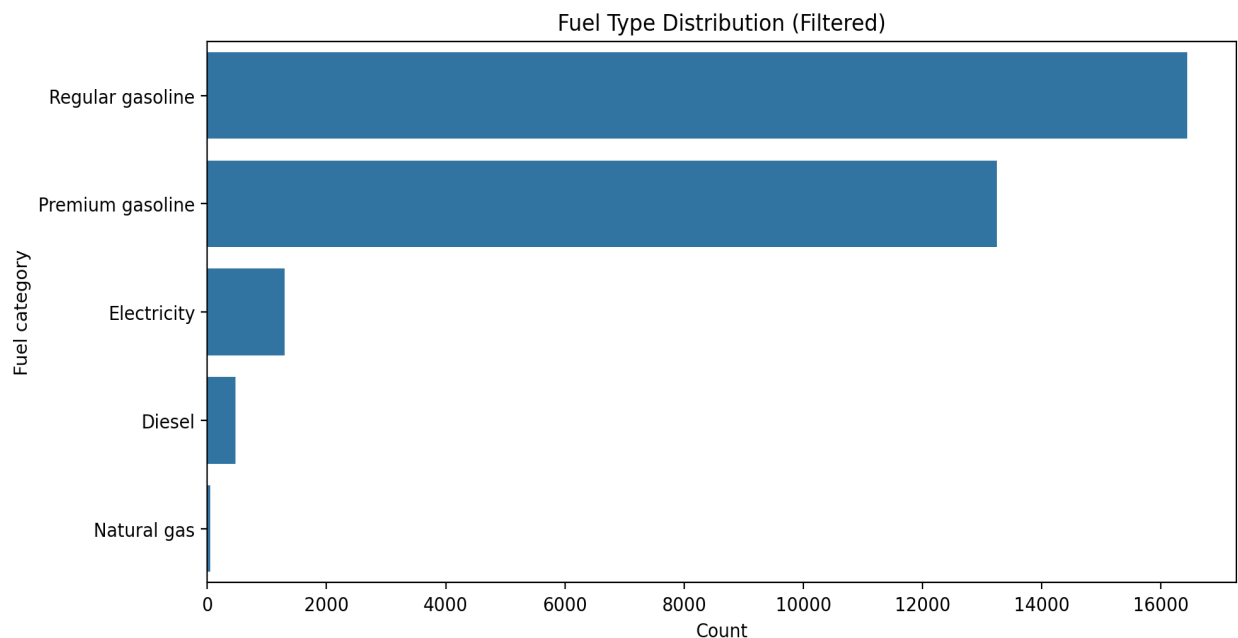


Figure: Counts of normalized fuel categories used in training/evaluation.

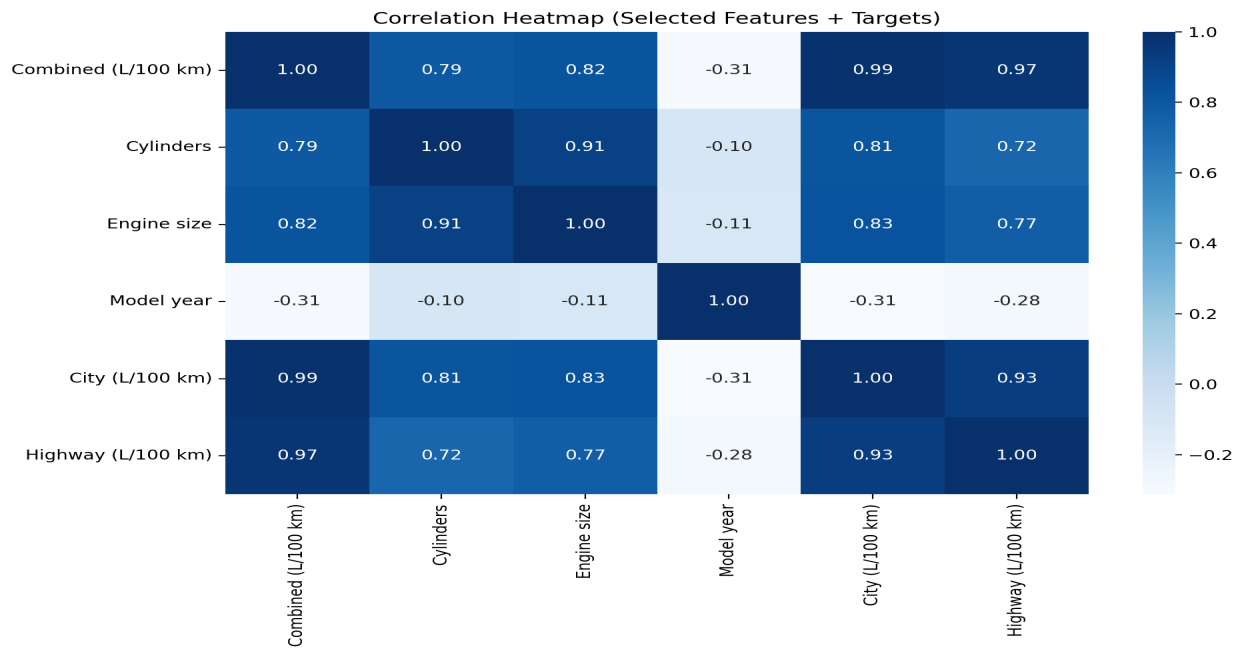


Figure: Correlation matrix for core numeric features and targets.

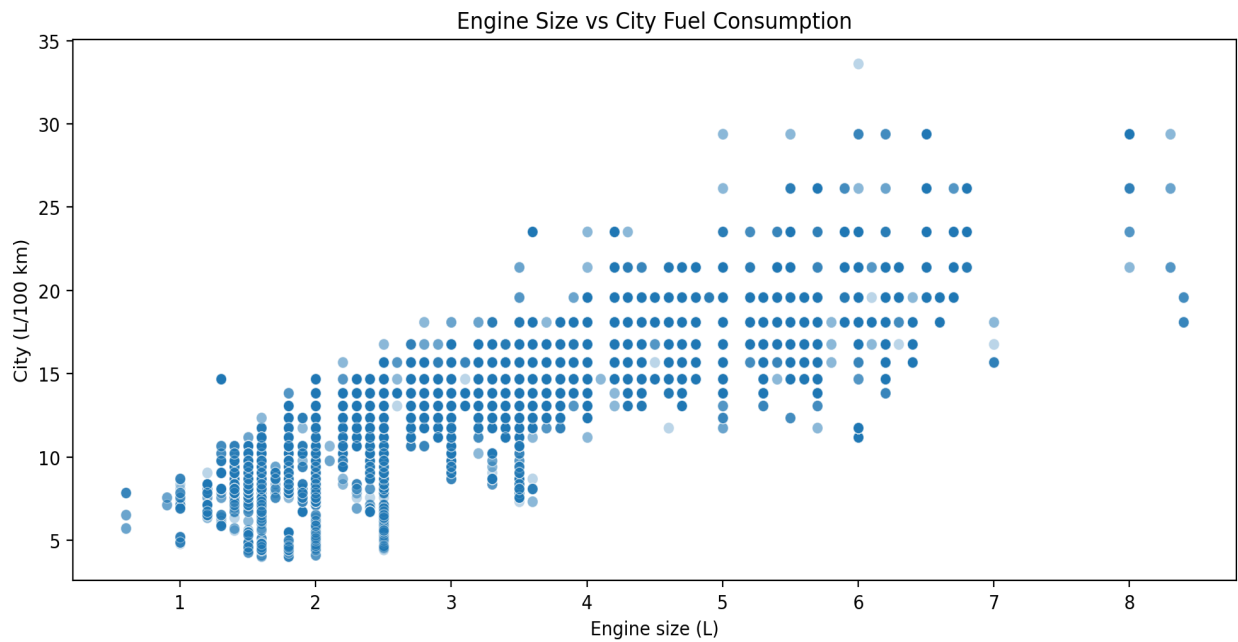


Figure: Relationship between displacement and city fuel consumption.

6. Machine Learning Model Development

6.1 Feature Set

The feature set mirrors the deployed application and includes: Combined consumption (L/100 km), Cylinders, Engine size (displacement), Model year, and one-hot encoded fuel categories.

6.2 Models

The deployed system is an ensemble-style comparison of multiple regressors trained to predict City and Highway fuel consumption jointly. Models are loaded from serialized .pkl files and evaluated consistently on the same held-out test split.

Models evaluated (when available): Random Forest, XGBoost, LightGBM, Decision Tree, and CatBoost.

7. Model Evaluation and Results

Metrics reported: MAE, RMSE, and R^2 for each target plus their averages. Lower MAE/RMSE is better; higher R^2 is better.

Model	MAE City	RMSE City	R2 City	MAE Highway	RMSE Highway	R2 Highway	RMSE Avg
CatBoost	0.606	0.880	0.936	0.687	0.963	0.823	0.921
Decision Tree	1.458	1.878	0.709	0.762	0.997	0.810	1.437
Random Forest	1.203	1.369	0.845	1.853	2.061	0.187	1.715
Ensemble (avg)	2.330	2.528	0.473	0.788	1.006	0.806	1.767
XGBoost	0.770	1.022	0.914	3.176	3.689	-1.603	2.356
LightGBM	9.163	9.790	-6.907	4.628	5.170	-4.112	7.480

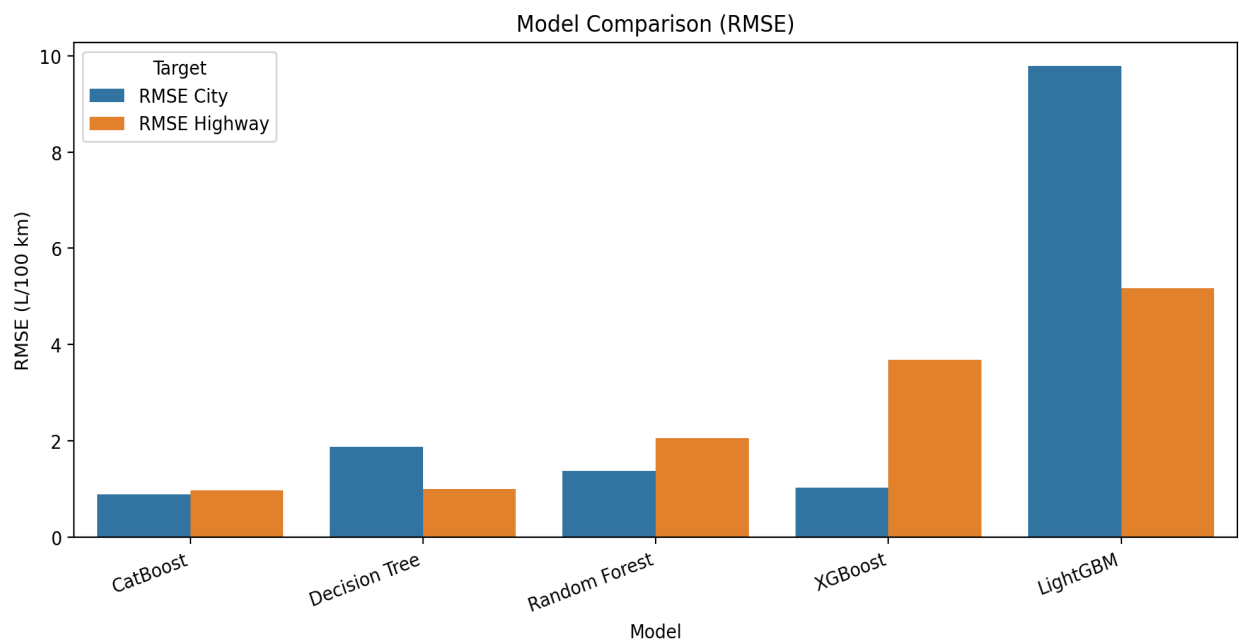


Figure: RMSE comparison for City and Highway targets across models.

8. Conclusions and Future Work

This report demonstrates an end-to-end workflow from dataset preparation and EDA to model evaluation and deployment. Future work can include richer feature engineering (e.g., vehicle class, transmission descriptors) and improved robustness across new model years.

Deployment

The final models are exposed via a Streamlit web application that collects the same feature set and returns City and Highway predictions in L/100 km.

Live app: <https://fuel-consumption-calc.streamlit.app/>

Appendix: Notebook Notes (Markdown Cells)

The following section includes markdown notes extracted from the referenced notebook (best-effort).

Task

LGBM, XGBoost, and Random Forest models will be initialized, trained, and evaluated using the scaled training data (X_train_scaled, y_train).

Train LGBM Model

Subtask:

Initialize, train, and evaluate a LightGBM Regressor model on the scaled training data (X_train_scaled, y_train).

Reasoning:

The subtask requires initializing, training, and evaluating a LightGBM Regressor model. This involves importing the necessary class, fitting the model to the scaled training data, making predictions on the scaled test data, and then calculating the Mean Squared Error, Root Mean Squared Error, and R-squared to evaluate its performance.

Reasoning:

Following the pattern of evaluating previous models, I will now evaluate the LightGBM model on the training set to compare its performance with the test set performance.

Reasoning:

The next logical step in the task is to initialize, train, and evaluate an XGBoost Regressor model, similar to how the LightGBM model was handled.

Reasoning:

Following the pattern of evaluating previous models, I will now evaluate the XGBoost model on the training set to compare its performance with the test set performance.

Reasoning:

The final step in the main task is to initialize, train, and evaluate a Random Forest Regressor model, following the same evaluation process used for the LGBM and XGBoost models.

Reasoning:

Following the pattern of evaluating previous models, I will now evaluate the Random Forest model on the training set to compare its performance with the test set performance.

Reasoning:

Next, I will initialize, train, evaluate, and save a CatBoost Regressor model. This involves importing the necessary classes, fitting the model to the training data, making predictions, calculating evaluation metrics (MSE, RMSE, R-squared) for both test and training sets, and finally saving the trained model to a file.